Lecture 8 Practice Questions

Question 1 (q1.py)

Make a Python version of the math function f(x) = 2x + 1. Use it print the values f(2), f(5), and f(1000). Then ask the user to enter a number and print the value of f(x) as shown.

```
5
11
2001
Enter a number: 2.6
f(2.6) = 6.2
```

Question 2 (q2.py)

Write a function that takes a person's name as input, and returns a string of the form 'Hello, <name>!'. Then ask the user to enter the names of two people, and print the result of calling the function for each person.

```
Person 1: Audrey
Person 2: Ali

Hello, Audrey!
Hello, Ali!
```

Question 3 (q3.py)

Write a function called print_in_bars(word) that prints word surrounded by | characters as shown. This function only prints, and so has no return. Then ask the user to enter their three favourite foods, and use print_in_bars to print each food.

```
Food 1: peaches
Food 2: sushi
Food 3: ice cream

|peaches|
|sushi|
|ice cream|
```

Question 4 (q4.py)

Write a function called print_in_box(word) that prints word in a box as shown. Use the print_in_bars function from the previous question to print the middle line of the box. Then ask the user to enter a movie name and print in a box.

```
Enter a movie name: The Matrix
+----------+
|The Matrix|
+----------+
```

Question 5 (q5.py)

Write a program that asks the user to enter an int that is 0 or greater, and a word, and then prints the word in a numbered list as shown. Use a for-loop in your solution. No functions are required, but you can make one if you want.

```
Enter an int: 3
Enter a word: hello!
hello!
hello!
hello!
```

Question 6 (q6.py)

Write a program that asks the user to enter an int (called n)that is 0 or greater. Then, starting at 1, print the ints from 1 to n on the screen in a box as shown. Use the print_in_box(word) function from the previous question.

```
Enter an int: 3

+-+
|1|
+-+

+-+
|2|
+-+

+-+
|3|
+-+
```